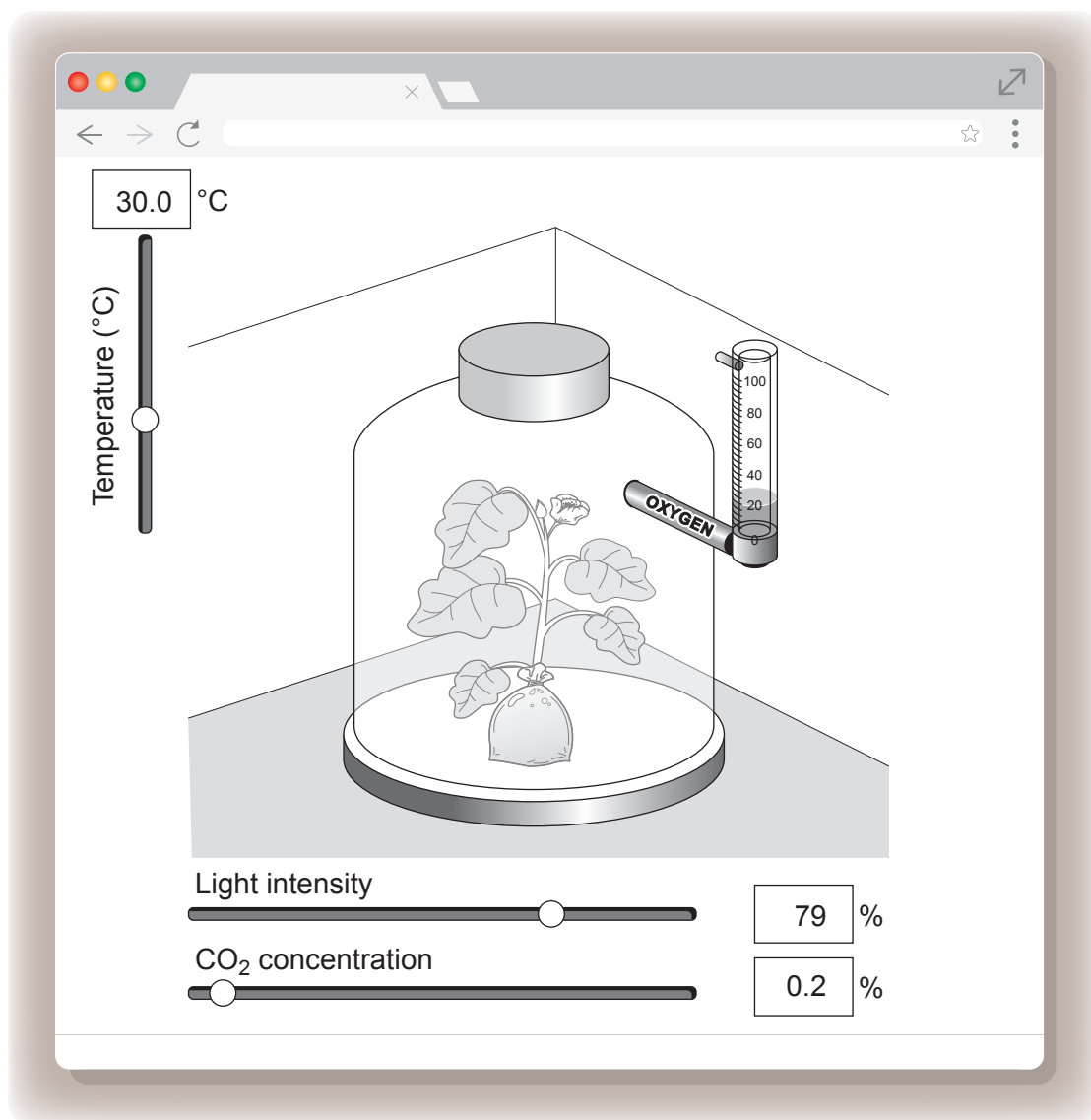


4. (a) State the word equation for photosynthesis.

[2]

- (b) Students used a computer simulation to investigate how the limiting factors of photosynthesis affect its rate. A screenshot from the simulation is shown in **Image 4.1**.

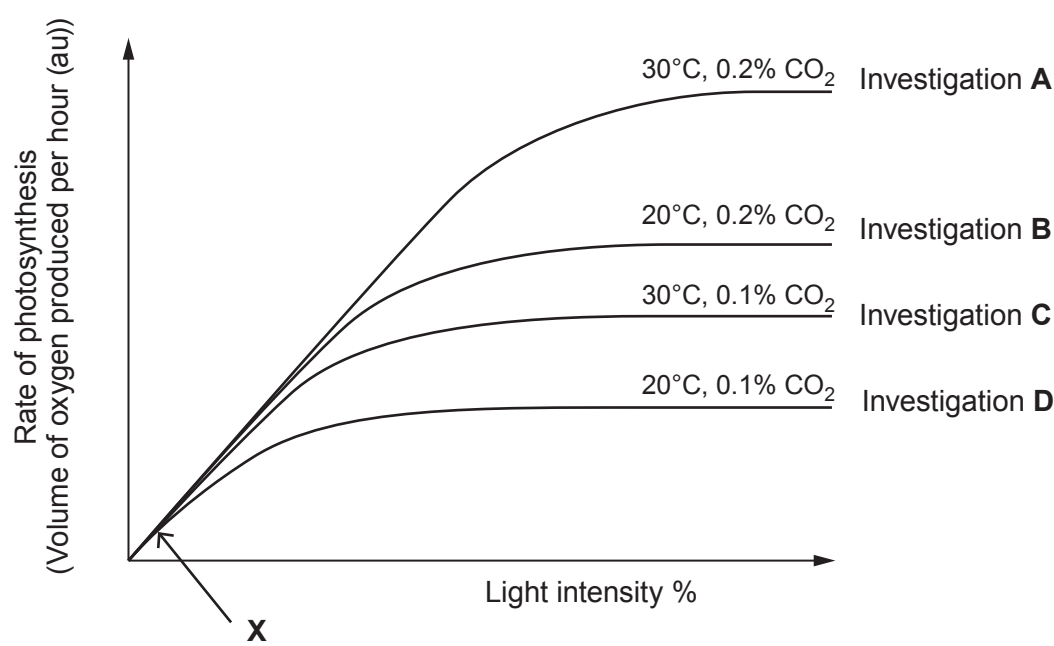
Image 4.1



The simulation allowed the students to carry out investigations into the rate of photosynthesis by adjusting the temperature, light intensity and carbon dioxide concentration the plant received. The rate of photosynthesis was then calculated by measuring the volume of **oxygen** produced by the plant per hour. The computer then produced a graph of the results as shown in **Graph 4.2**.



Graph 4.2



(i) With reference to limiting factors, explain why the rate of photosynthesis is:

I. lower in investigation C than in investigation A;

[2]

.....

.....

.....

.....

II. higher in investigation C than in investigation D.

[2]

.....

.....

.....

.....

(ii) Suggest how you could increase the rate of photosynthesis in investigation A.

[1]

.....

.....

(iii) State the limiting factor which is controlling the rate of photosynthesis at point X on Graph 4.2.

[1]

.....



Examiner
only

(iv) The levels were adjusted to the following:

- light intensity to bright daylight levels
- CO₂ concentration to 0.04%
- temperature to 80 °C.

I. Suggest how a temperature of 80 °C would affect the rate of photosynthesis and explain your answer. [2]

.....

.....

.....

.....

II. Suggest why a CO₂ concentration of 0.04% was chosen. [1]

.....

.....

(c) Explain why the volume of oxygen produced by the plant per hour can be used as a measurement of the rate of photosynthesis. [1]

.....

.....

.....

.....

12

